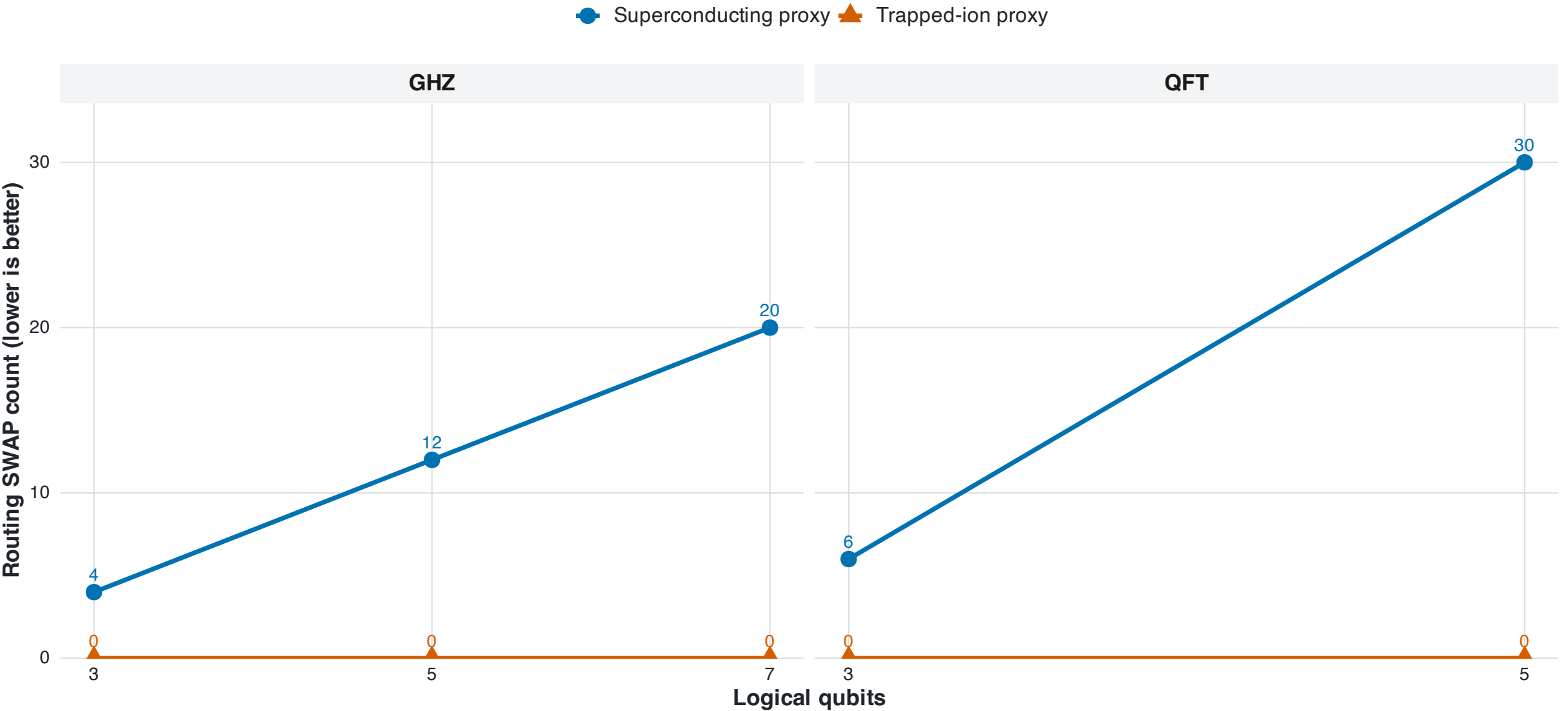


Long-Range Circuits Magnify the Connectivity Gap

Routing SWAPs grow with circuit size on the line-connected proxy



Evidence class: offline architecture proxy. Lines connect tested sizes; they are not extrapolations.

- QUESTION — What happens as GHZ and QFT circuits become larger?
- READ — Move left to right as qubit count grows; a rising line means routing overhead is increasing.
- RESULT — The superconducting line proxy reaches 20 SWAPs for GHZ-7 and 30 for QFT-5; the all-to-all proxy stays at 0 for these tests.
- WHY IT MATTERS — Algorithms with distant two-qubit interactions are especially sensitive to device connectivity.
- BOUNDARY — Only the configured 3–7 qubit cases were tested; the graph does not predict every larger circuit.